

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – Pseudocode Design

|                         |                                                                                                                                                                                                               |                      |                                            |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------|
| <b>Course Code:</b>     | <b>DSA0405</b>                                                                                                                                                                                                | <b>Course Title:</b> | <b>FUNDAMENTALS OF DATA SCIENCE</b>        |
| <b>CO Assessed:</b>     | <b>CO5 –Pseudocode Design</b>                                                                                                                                                                                 | <b>Assessment:</b>   | <b>Assessment Tool 1-Pseudocode Design</b> |
| <b>Weightage:</b>       | <b>30% of CIA</b>                                                                                                                                                                                             | <b>Total Marks:</b>  | <b>100</b>                                 |
| <b>CO5 Statement:</b>   | Develop and analyze machine learning models using linear regression, logistic regression, and unsupervised learning techniques for prediction, classification, model evaluation, and pattern discovery. (BL4) |                      |                                            |
| <b>Student Name:</b>    | Y. Jayesh Ranjan                                                                                                                                                                                              | <b>Reg. No.:</b>     | 192472201                                  |
| <b>Section / Batch:</b> | CSE – AI                                                                                                                                                                                                      | <b>Date:</b>         | 25/08/26                                   |

### Code of Conduct

I Y. Jayesh Ranjan – 192472201 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Jayesh Ranjan

| <b>Criteria</b>                | <b>Weight age</b> | <b>Level 4 – Exemplary</b>                                              | <b>Level 3 – Proficient</b>                           | <b>Level 2 – Developing</b>                    | <b>Level 1 – Beginning</b>                                  | <b>Marks</b> |
|--------------------------------|-------------------|-------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|-------------------------------------------------------------|--------------|
| <b>Understanding of Case</b>   | <b>25</b>         | <b>Thorough understanding; clearly identifies all key issues</b>        | <b>Good understanding with most issues identified</b> | <b>Basic understanding; some issues missed</b> | <b>Poor understanding; problem not identified correctly</b> |              |
| <b>Application of Concepts</b> | <b>25</b>         | <b>Effectively applies relevant concepts to analyze the case deeply</b> | <b>Applies appropriate concepts with minor gaps</b>   | <b>Limited application of concepts</b>         | <b>Fails to apply relevant concepts</b>                     |              |
| <b>Analysis</b>                | <b>20</b>         | <b>In-depth analysis with strong reasoning and insights</b>             | <b>Good analysis with logical reasoning</b>           | <b>Basic analysis with limited depth</b>       | <b>Weak or no analysis; lacks reasoning</b>                 |              |

|                        |           |                                                                 |                                                      |                                                |                                              |  |
|------------------------|-----------|-----------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|----------------------------------------------|--|
| <b>Solution Design</b> | <b>20</b> | <b>Innovative, feasible solutions with strong justification</b> | <b>Logical solutions with adequate justification</b> | <b>Basic solutions with weak justification</b> | <b>Incorrect or no solution provided</b>     |  |
| <b>Clarity</b>         | <b>10</b> | <b>Clear, well-structured, and easy to understand</b>           | <b>Organized and understandable presentation</b>     | <b>Limited clarity and structure</b>           | <b>Poor clarity; difficult to understand</b> |  |

## QUESTION 11: LINEAR REGRESSION – MSE AND R<sup>2</sup> SCORE

### Problem Statement

Linear Regression is a supervised machine learning technique used to predict a continuous numerical value from one or more input features. In this problem, a Linear Regression model is trained using a given dataset containing independent variables and a dependent variable. The dataset is divided into training data and testing data so that the performance of the trained model can be evaluated. The model learns the relationship between the input features and the target value from the training data. After training, the testing data is provided to the model to obtain predicted values. The difference between the actual and predicted values is called the residual. Mean Squared Error (MSE) is calculated to measure the average squared prediction error. Finally, the R<sup>2</sup> score is calculated to determine how well the model explains the variation in the target values.

### Algorithm

1. Start the Linear Regression model evaluation process.
2. Load the dataset containing the independent variables and dependent target variable.
3. Identify and select the input features required for prediction.
4. Identify the target variable that the model has to predict.
5. Separate the input features and target values into X and Y.
6. Divide the dataset into training data and testing data.
7. Create a Linear Regression model for learning the relationship between X and Y.
8. Train the Linear Regression model using the training dataset.
9. Provide the testing input data to the trained model.
10. Calculate the predicted target values for the testing dataset.
11. Calculate the residual for each observation using actual value minus predicted value.
12. Calculate the Mean Squared Error by taking the average of squared residuals.
13. Calculate the R<sup>2</sup> score using the actual and predicted target values.

14. Display the predicted values and residual values obtained from the model.
15. Display the MSE and  $R^2$  score as the final evaluation results.
16. Stop the process.

## Pseudocode

BEGIN

LOAD dataset

SELECT input features

STORE as X

SELECT target variable

STORE as Y

SPLIT X and Y into

Training Data

Testing Data

CREATE Linear Regression model

TRAIN model using Training Data

PREDICT target values using Testing Data

STORE predicted values as Y\_PRED

FOR each test observation

Residual = Actual Value – Predicted Value

END FOR

Calculate MSE

MSE = Average of

$(\text{Actual Value} - \text{Predicted Value})^2$

Calculate  $R^2$  Score

DISPLAY Predicted Values

DISPLAY Residuals

DISPLAY MSE

DISPLAY  $R^2$  Score

END

## Sample Output

Linear Regression Model

Predicted Values:

[52.4, 61.8, 70.5, 78.2, 85.6]

Residuals:

[1.6, -1.8, 2.5, -2.2, 1.4]

Mean Squared Error (MSE): 4.05

R<sup>2</sup> Score: 0.94

Model evaluation completed successfully.

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## **QUESTION 12: LOGISTIC REGRESSION – STUDENT PASS OR FAIL**

### **Problem Statement**

Logistic Regression is a supervised machine learning algorithm used mainly for classification problems. In this problem, Logistic Regression is used to predict whether a student will Pass or Fail based on study hours and attendance percentage. The student dataset contains study hours, attendance percentage, and the corresponding result of the student. Study hours and attendance percentage are considered input features, while Pass or Fail is considered the target variable. The dataset is divided into training and testing data to train and evaluate the classification model. After training, the model receives the study hours and attendance percentage of a new student. It calculates the probability that the student will pass. A threshold value of 0.5 is then used to convert the probability into the final Pass or Fail classification.

### **Algorithm**

1. Start the student result prediction process.
2. Load the student dataset containing study hours, attendance percentage, and results.
3. Select study hours as the first input feature.
4. Select attendance percentage as the second input feature.
5. Select Pass or Fail as the target variable.
6. Separate the input features and target variable into X and Y.
7. Divide the dataset into training and testing data.
8. Create a Logistic Regression classification model.
9. Train the model using the available training data.
10. Take the study hours of a new student as input.
11. Take the attendance percentage of the new student as input.
12. Use the trained model to calculate the probability of the student passing.
13. Compare the calculated probability with the threshold value 0.5.
14. If the probability is greater than or equal to 0.5, classify the student as Pass.
15. If the probability is less than 0.5, classify the student as Fail.

## Pseudocode

```
BEGIN
    LOAD student dataset
    SELECT Study Hours
    SELECT Attendance Percentage
    STORE both as X
    SELECT Student Result
    STORE as Y
    SPLIT dataset into
        Training Data
        Testing Data

    IF probability  $\geq$  0.5 THEN
        Result = "PASS"
    ELSE
        Result = "FAIL"
    END IF
    DISPLAY probability
    DISPLAY Result
END
```

## Sample Output

Student Details  
Study Hours: 6 hours  
Attendance Percentage: 85%  
Predicted Probability of Pass: 0.87  
Predicted Result: PASS  
Student classification completed successfully.

## **QUESTION 13: LOGISTIC REGRESSION – SIGMOID PROBABILITY**

### **Problem Statement**

Logistic Regression uses the sigmoid function to convert a numerical value into a probability between 0 and 1. In this problem, the input feature values, weights, and bias are used to calculate a linear value represented by  $z$ . The calculated  $z$  value is given as input to the sigmoid function. The sigmoid function produces a probability value that represents the likelihood of an observation belonging to class 1. A threshold value of 0.5 is used to convert the probability into a binary class. If the probability is greater than or equal to 0.5, the observation is classified as class 1. If the probability is less than 0.5, the observation is classified as class 0. Finally, the calculated probability and binary class are displayed as the output.

### **Algorithm**

1. Start the sigmoid probability calculation process.
2. Input the required feature values for the classification problem.
3. Initialize the weights corresponding to the input features.
4. Initialize the bias value used in the logistic regression equation.
5. Calculate the linear value  $z$  using the input features, weights, and bias.
6. Apply the sigmoid function to the calculated  $z$  value.
7. Calculate the probability using the sigmoid formula.
8. Store the calculated value as the probability of class 1.
9. Compare the probability with the threshold value 0.5.
10. If the probability is greater than or equal to 0.5, assign class 1.
11. If the probability is less than 0.5, assign class 0.
12. Display the calculated sigmoid probability.
13. Display the final binary classification result.
14. Stop the process.



## Pseudocode

BEGIN

INPUT feature value X

INPUT weight W

INPUT bias B

Calculate:

$$z = B + (W \times X)$$

Calculate sigmoid probability:

$$P = 1 / (1 + \text{EXP}(-z))$$

IF  $P \geq 0.5$  THEN

Class = 1

ELSE

Class = 0

END IF

DISPLAY Probability P

DISPLAY Class

END

## Sample Output

Input Feature Value: 2.5

Weight: 1.2

Bias: -1.0

Calculated z value: 2.0

Sigmoid Probability: 0.8808

Since Probability  $\geq 0.5$

Predicted Class: 1

## QUESTION 14: LOGISTIC REGRESSION – EVALUATION METRICS

### Problem Statement

A Logistic Regression classifier must be evaluated to determine how accurately it predicts the classes of new observations. In this problem, the actual class labels and predicted class labels are compared to create a confusion matrix. The confusion matrix contains four important values called True Positive, True Negative, False Positive, and False Negative. These values are used to calculate different performance measures for the classification model. Accuracy measures the overall percentage of correct predictions made by the classifier. Precision measures the correctness of positive predictions, while recall measures the ability to correctly identify actual positive cases. Finally, the F1-score combines precision and recall into a single performance measure. The calculated values are displayed to evaluate the overall performance of the Logistic Regression model.

### Algorithm

1. Start the Logistic Regression evaluation process.
2. Obtain the actual class labels from the testing dataset.
3. Obtain the predicted class labels generated by the Logistic Regression model.
4. Initialize TP, TN, FP, and FN values to zero.
5. Compare each actual label with its corresponding predicted label.
6. If actual and predicted values are both 1, increase the True Positive count.
7. If actual and predicted values are both 0, increase the True Negative count.
8. If actual is 0 and predicted is 1, increase the False Positive count.
9. If actual is 1 and predicted is 0, increase the False Negative count.
10. Calculate accuracy using TP, TN, FP, and FN values.
11. Calculate precision using the True Positive and False Positive values.
12. Calculate recall using the True Positive and False Negative values.
13. Calculate F1-score using the calculated precision and recall values.
14. Display the confusion matrix values.

15. Display accuracy, precision, recall, and F1-score.

16. Stop the evaluation process.

## Pseudocode

BEGIN

INPUT Actual\_Labels

INPUT Predicted\_Labels

SET TP = 0

SET TN = 0

SET FP = 0

SET FN = 0

FOR each observation

IF Actual = 1 AND Predicted = 1 THEN

TP = TP + 1

ELSE IF Actual = 0 AND Predicted = 0 THEN

TN = TN + 1

ELSE IF Actual = 0 AND Predicted = 1 THEN

FP = FP + 1

ELSE IF Actual = 1 AND Predicted = 0 THEN

FN = FN + 1

END IF

END FOR

Accuracy =

$(TP + TN) /$

$(TP + TN + FP + FN)$

Precision =

$TP / (TP + FP)$

Recall =

$$TP / (TP + FN)$$

F1\_Score =

$$2 \times \text{Precision} \times \text{Recall} /$$
$$(\text{Precision} + \text{Recall})$$

DISPLAY TP

DISPLAY TN

DISPLAY FP

DISPLAY FN

DISPLAY Accuracy

DISPLAY Precision

DISPLAY Recall

DISPLAY F1\_Score

END

## Sample Output

Confusion Matrix Values

True Positive (TP): 45

True Negative (TN): 40

False Positive (FP): 5

False Negative (FN): 10

Accuracy: 0.85

Precision: 0.90

Recall: 0.82

F1-Score: 0.86

## QUESTION 15: UNSUPERVISED LEARNING

### Problem Statement

Unsupervised Learning is a machine learning approach in which the dataset does not contain predefined target or class labels. In this problem, an unlabelled dataset is provided to the learning system to discover hidden patterns and natural groups within the data. Since there is no target variable, the system works only with the available input features. First, the dataset is loaded and the required features are selected for analysis. The selected features are then preprocessed by checking missing values and scaling or normalizing the data when necessary. After preprocessing, an appropriate clustering algorithm is applied to identify similar groups of observations. Each observation is assigned to a suitable cluster based on its characteristics. Finally, the discovered clusters and cluster assignments are displayed as the output.

### Algorithm

1. Start the unsupervised learning process.
2. Load the unlabelled dataset into the system for analysis.
3. Select the required input features from the dataset.
4. Check the selected features for missing or invalid values.
5. Handle missing values using an appropriate preprocessing method if required.
6. Convert the selected features into a suitable format for processing.
7. Scale or normalize the features when they have different ranges.
8. Select a suitable unsupervised learning or clustering algorithm.
9. Apply the selected algorithm to the preprocessed dataset.
10. Analyze the similarities and differences between the observations.
11. Identify the natural groups or clusters present in the dataset.
12. Assign each observation to its appropriate cluster.
13. Store the cluster number assigned to every observation.
14. Display the discovered clusters and their corresponding observations.
15. Verify that every observation has been assigned to a cluster.
16. Stop the unsupervised learning process.

## Pseudocode

```
BEGIN
  LOAD unlabelled dataset
  SELECT required features
  CHECK for missing values
  IF missing values exist THEN
    HANDLE missing values
  END IF
  PREPROCESS selected features
  SCALE or NORMALIZE the features
  SELECT clustering algorithm
  APPLY clustering algorithm
  to the preprocessed data
  FOR each data point
    ANALYZE data point
    FIND appropriate cluster
    ASSIGN data point
    to the cluster
  END FOR
  STORE cluster assignments
  DISPLAY discovered clusters
  DISPLAY cluster assignment
  for each data point
END
```

## Sample Output

Unsupervised Learning Process

Number of Data Points: 10

Number of Features: 2

Preprocessing completed successfully.

Clusters Discovered: 3

Cluster Assignments:

Data Point 1 → Cluster 1

Data Point 2 → Cluster 1

Data Point 3 → Cluster 2

Data Point 4 → Cluster 2

Data Point 5 → Cluster 3

Data Point 6 → Cluster 1

Data Point 7 → Cluster 3

Data Point 8 → Cluster 2

Data Point 9 → Cluster 3

Data Point 10 → Cluster 1

Clustering completed successfully.